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English for Computer Science

Neural Networks

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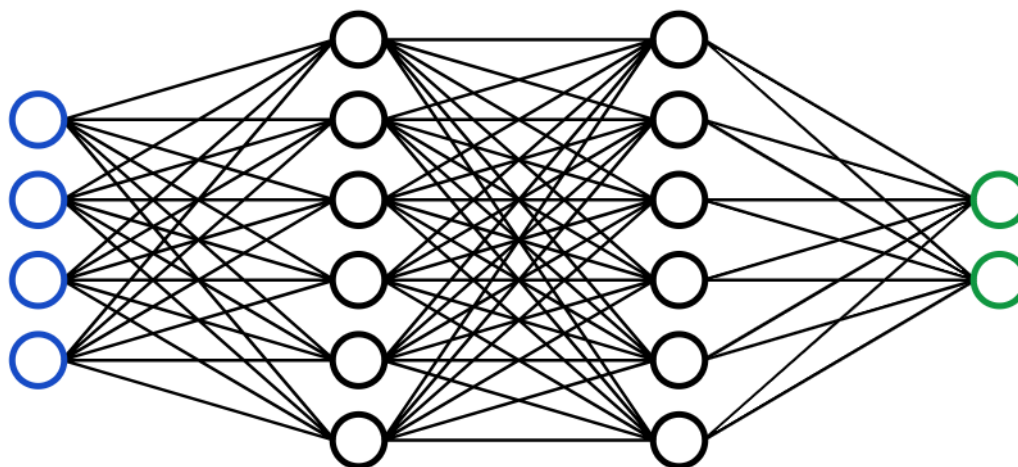
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Chapter 1

Introduction

1.1 What is a neural network?

Neural networks are a method in artificial intelligence inspired by the human brain. They fall under a type of machine learning called deep learning. Neural networks use interconnected nodes or neurons in a layered structure. This structure includes an input layer, hidden layers, and an output layer, resembling the way the human brain processes information.



The learning process involves feeding data into the network and adjusting the connections based on the output error. This is done through a method called backpropagation, which adjusts the weights of the connections to minimize errors. This iterative process allows the network to learn from its mistakes and improve continuously.

Neural networks are capable of solving complex problems with high accuracy. They excel in various applications such as image recognition, where they identify and classify objects in images; natural language processing, where they understand and generate human language; speech recognition, converting spoken language into text; and game playing, where they can outperform human players in strategic games.

Neural networks provide a powerful tool for creating adaptive systems that improve over time, enabling significant advancements in technology and artificial intelligence.

1.2 What are Neural networks used for?

Neural networks can help computers make intelligent decisions with limited human assistance because they can learn and model the relationships between input and output data that are nonlinear and complex. They can make generalizations and inferences.

Neural networks have several use cases across many industries. These are four of the important applications of neural networks:

- Computer vision: the ability of computers to extract information and insights from images and videos. With neural networks, computers can distinguish and recognize images similar to humans. Computer vision has several applications, such as:
 - Medical diagnosis by medical image classification
 - Facial recognition to identify faces and recognize attributes like open eyes, glasses, and facial hair
 - Image labeling to identify brand logos, clothing, safety gear, and other image details
- Speech recognition: Neural networks can analyze human speech despite varying speech patterns, pitch, tone, language, and accent. Virtual assistants like Amazon Alexa and automatic transcription software use speech recognition to do tasks like:
 - Convert clinical conversations into documentation in real time
 - Accurately subtitle videos and meeting recordings for wider content reach
- Natural language processing (NLP): the ability of computers to process natural, human-created text. Neural networks help computers gather insights and meaning from text data and documents. NLP has several use cases, including in these functions:
 - Automated virtual agents and chatbots
 - Automatic organization and classification of written data

- Document summarization and article generation for a given topic
- Recommendation engines: Neural networks can track user activity to develop personalized recommendations. They can also analyze all user behavior and discover new products or services that interest a specific user.

Chapter 2

Neural network mechanism

2.1 Neurons

The human visual system is incredibly sensitive, with the ability to recognize small details in the surrounding environment.

For example, when you're walking in a flower garden, you can distinguish between different types of flowers, identify their colors, shapes, and structures. The human eye is capable of discerning tiny details such as color spots, lines, or intricate structures on the petals.

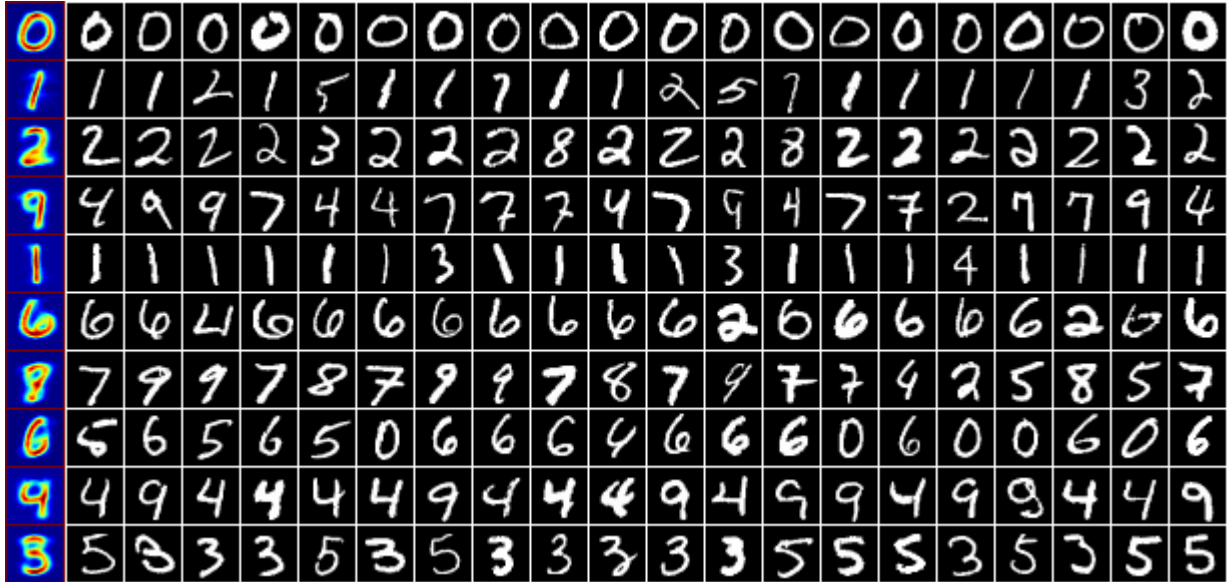
Another example, consider the following handwritten digit sequence:

A handwritten digit sequence '504192' in a cursive, slightly blurred style. The digits are black on a white background. The '5' is a simple vertical stroke with a small hook. The '0' is a circle with a small tail. The '4' is a vertical stroke with a small hook. The '1' is a simple vertical stroke. The '9' is a vertical stroke with a small hook. The '2' is a vertical stroke with a small hook.

Most people would easily recognize those digits as 504192.

In fact, the sharpness of our eyesight is thanks to the human brain. The brain contains millions of visual nerve cells, with connections between them to carry out image processing, whether simple or complex. We carry a supercomputer in our heads.

As in the handwriting recognition example above, it's a task that's far from easy. Yet, we often don't appreciate the level of difficulty our visual system handles. And that difficulty becomes apparent when you attempt to write a computer program to recognize digits. Then you'll be drawn into countless edge cases.



Neural networks approach the problem of handwriting recognition by using a large number of handwritten digits as training examples. By developing a system capable of learning from these examples, neural networks can automatically infer the rules for recognizing handwritten digits. This allows neural networks to improve their accuracy by increasing the number of training samples. Neural networks can learn from a vast amount of data and enhance their recognition capabilities through the process of experiential learning.

For example, the network starts with a number of neurons corresponding to each pixel in the 28x28 input image, totaling 784 neurons. Each of these holds a number representing the grayscale value of the corresponding pixel, ranging from 0 for black pixels to 1 for white pixels. The number inside these neurons is called their activation, and the image you can think of here is that each neuron will light up when its activation is a large number.

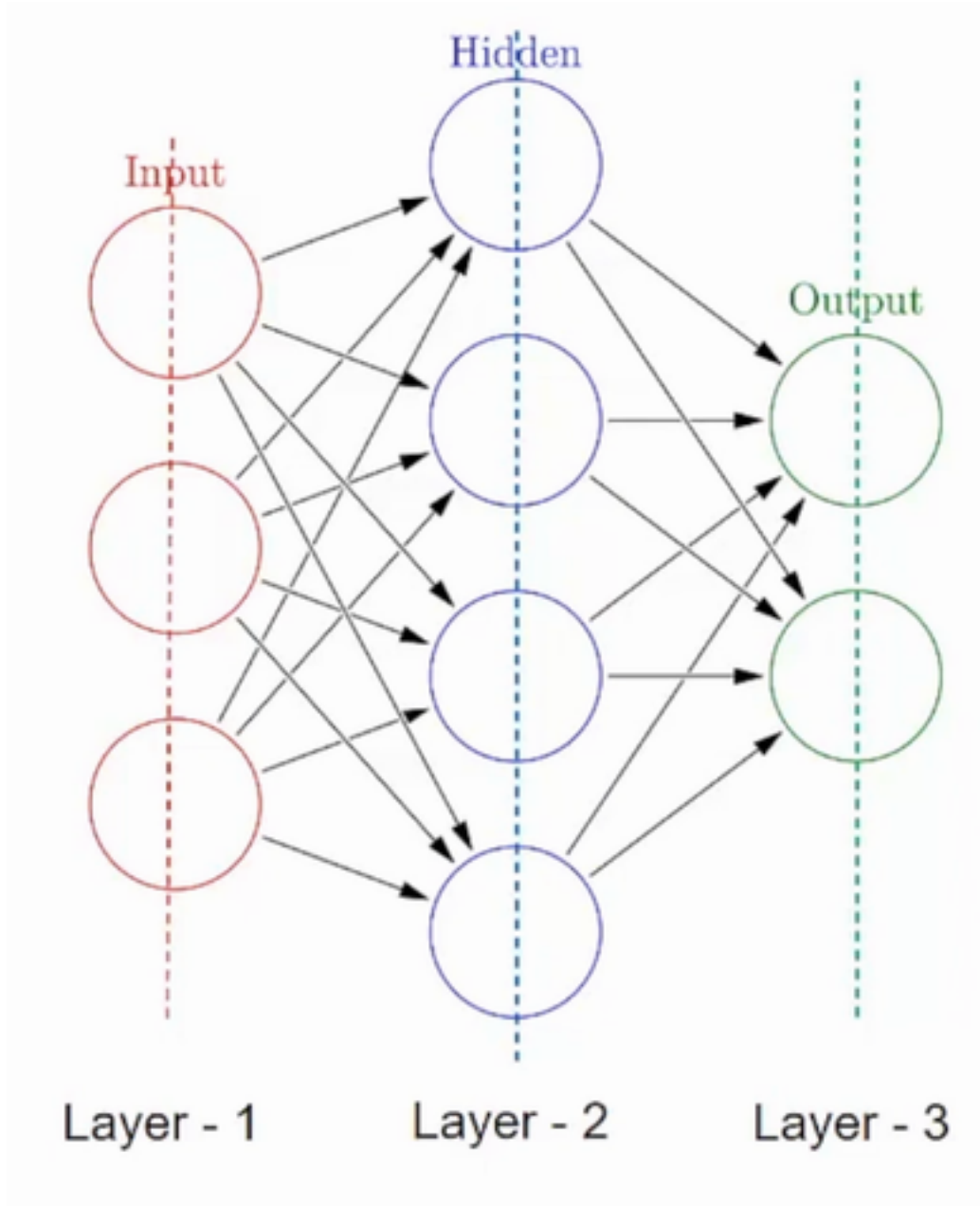
2.2 Structure

Artificial neural networks replicate this structure with artificial neurons, or nodes, collaborating to tackle problems. These artificial neurons are software components within algorithms or programs, utilizing computing systems to perform mathematical computations at their essence.

A basic neural network has interconnected artificial neurons organized into three layers:

- **Input Layer:** This is where information from the outside world enters the artificial neural network.

- Hidden Layer: Hidden layers take their input from the input layer or other hidden layers. Artificial neural networks can have a large number of hidden layers. Each hidden layer analyzes the output from the previous layer, processes it further, and passes it on to the next layer.
- Output Layer: The output layer gives the final result of all the data processing by the artificial neural network.



For example: Let's assume we have a neural network to recognize handwritten digits from 28x28 pixel images.

- The input layer would have 784 neurons ($28 \times 28 = 784$).
- If we have two hidden layers, each with 16 neurons, each neuron in the first hidden layer will receive signals from all 784 neurons in the input layer, process the information, and then send signals to the neurons in the second hidden layer. The second hidden layer continues processing and transmits the results to the output layer.
- The output layer has 10 neurons, each representing a digit from 0 to 9. The neuron with the highest value (closest to 1) indicates the recognized digit from the image.

2.3 Functioning

Neural networks operate through a series of interconnected neurons organized into layers. The functioning of NNs can be broadly categorized into two main processes: Feedforward and Backpropagation.

2.3.1 Feedforward Process

The feedforward process is the initial phase where input data is processed through the network to generate an output. Here's a step-by-step breakdown:

- **Input Layer Reception:** The input layer receives raw data or features that need to be processed. Each neuron in the input layer corresponds to a feature or dimension of the input data.
- **Propagation Through Hidden Layers:** After receiving input, the data flows through one or more hidden layers. Each neuron in a hidden layer receives inputs from all neurons in the previous layer, weighted by connection weights.
- **Neuron Activation:** At each neuron, the weighted sum of inputs (including an optional bias term) is computed. This sum is then passed through an activation function. The activation function introduces non-linearity into the network, allowing it to learn complex patterns and relationships in the data.
- **Output Layer Calculation:** The final processed information emerges from the output layer. Neurons in the output layer combine the outputs from the last hidden layer, again weighted and activated, to produce the network's final prediction or output.
- **Prediction or Output:** The output layer provides the network's prediction or classification based on the input data. For instance, in image recognition, the output might indicate the probability of the input image belonging to different classes (e.g., cat, dog, car).

2.3.2 Backpropagation Process

Once the network generates an output, the backpropagation process begins. This phase adjusts the network's weights based on the error between the predicted output and the actual output. Here's how it works:

- **Error Calculation:** The predicted output from the feedforward pass is compared to the actual target output (supervised learning scenario) to calculate an error or loss function. Common loss functions include mean squared error (MSE) for regression tasks and categorical cross-entropy for classification tasks.
- **Gradient Descent:** Backpropagation uses gradient descent to minimize the error. It calculates the gradient of the loss function with respect to each weight in the network. The gradient indicates the direction and magnitude of adjustments needed to reduce the error. Adjustments are made to minimize the error through iterative updates.
- **Weight Update:** The weights of the network connections are updated in the opposite direction of the gradient (gradient descent), scaled by a learning rate parameter. This process continues iteratively, updating weights after each batch or individual data point (depending on the training strategy) until convergence criteria are met.
- **Propagation of Error:** The error is propagated backward through the network, layer by layer, adjusting weights based on their contribution to the error. Each layer's weights are updated based on the error gradient and the activation of neurons in the preceding layer.
- **Iterative Learning:** Feedforward and backpropagation processes are repeated iteratively over the entire dataset (epoch) to train the neural network. Training continues until the network achieves satisfactory performance or convergence.

2.4 Gradient descent

Gradient descent is a crucial method used for optimizing neural networks. In neural networks, its primary objective is to adjust parameters such as weights and biases to minimize the loss function, which measures the discrepancy between the network's predictions and the actual values in the training data. This process begins with computing the gradient of the loss function with respect to each parameter of the neural network using backpropagation.

- **Parameter Initialization:** Initially, the parameters of the neural network are randomly initialized or set to predefined values.
- **Gradient Calculation:** Next, the gradient of the loss function is computed by backpropagating from the output layer of the network to the input layer. This process leverages the chain rule to compute gradients for each parameter.

- **Parameter Update:** The network parameters are updated by moving a small step in the direction opposite to the gradient. This is performed with a predetermined learning rate to control the magnitude of the updates.
- **Iteration:** The gradient descent process iterates over each epoch (the entire training dataset) or through each batch (mini-batch gradient descent) until the loss function achieves the desired value or meets other stopping criteria.

Through these steps, gradient descent enables neural networks to learn from training data and adjust parameters so that the network can make more accurate predictions and classifications. This forms the foundation for successfully applying neural networks in various real-world applications, ranging from image recognition to natural language processing.

2.4.1 Loss Function

A loss function, also known as a cost function or objective function, plays a fundamental role in machine learning and neural networks. Its primary purpose is to quantify how well a model's predictions match the actual target values in the training data. The loss function essentially measures the discrepancy between predicted values and ground truth, providing a metric that guides the optimization process during training.

During training, the goal is to minimize the value of the loss function. This process involves adjusting the model parameters (weights and biases in neural networks) iteratively to reduce prediction errors and improve the model's accuracy.

The choice of loss function depends on the specific task and the nature of the data. Different types of problems (regression, classification) and data distributions (balanced, imbalanced) may require different loss functions.

For instance, mean squared error (MSE) is commonly used in regression tasks, while cross-entropy loss is prevalent in classification tasks, especially for multi-class classification.

2.4.2 Learning rate

The learning rate is a crucial parameter in gradient descent methods. It determines the speed and size of adjustments made to model parameters during training.

The learning rate is a small positive constant used to scale the gradient descent update step. It dictates the direction and magnitude of movements from the current parameter values towards the optimal solution in parameter space.

Impact on Learning Process:

- A high learning rate can cause large updates in parameter values, potentially overshooting the optimal point or making the learning process unstable (overshooting).
- A low learning rate slows down the convergence towards the optimal solution, prolonging training time.

Choosing an appropriate learning rate is often a trial-and-error process, evaluated based on model performance on validation datasets. Techniques such as automatic search and gradual decrease of the learning rate during training can be applied to improve model stability and performance.

2.5 Training

Neural network training is the process of teaching a neural network to perform a task. Neural networks learn by initially processing several large sets of labeled or unlabeled data. By using these examples, they can then process unknown inputs more accurately.

2.5.1 Supervised learning

During the training of ANN under supervised learning, data scientists give artificial neural networks labeled datasets that provide the right answer in advance. This means each training example is paired with an output label.

Each output is compared with the desired output that is labeled. An error signal is generated, if there is a difference between the actual output and the desired output. On the basis of this error signal, the weights are adjusted until the actual output is matched with the desired output.

2.5.2 Unsupervised Learning

This type of learning is done without the supervision of a teacher. This learning process is independent.

During the training of ANN under unsupervised learning, the network is trained on data without explicit labels. The goal is to discover underlying patterns or structures in the data.

There is no feedback from the environment as to what should be the desired output and if it is correct or incorrect. Hence, in this type of learning, the network itself must discover the patterns and features from the input data, and the relation for the input data over the output.

2.5.3 Reinforcement Learning

This type of learning is used to reinforce or strengthen the network over some critic information. This learning process is similar to supervised learning, however we might have very less information.

During the training of network under reinforcement learning, the network receives some feedback from the environment. This makes it somewhat similar to supervised learning. However, the feedback obtained here is evaluative not instructive, which means there is no teacher as in supervised learning. After receiving the feedback, the network performs adjustments of the weights to get better critic information in future.

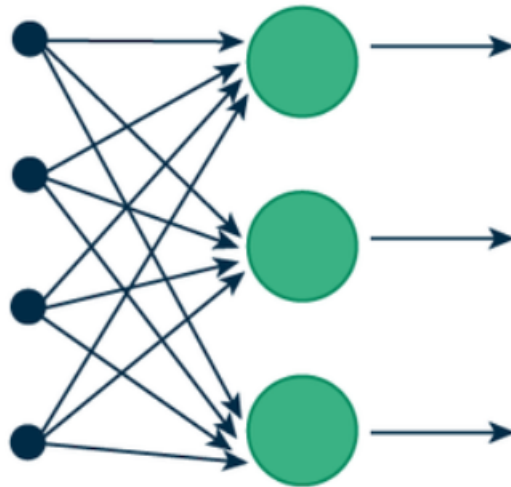
Chapter 3

Types of neural networks

Neural networks can be classified based on their architecture, purpose, and the type of data they process. Here are the main types, their definitions, and key characteristics

3.1 Feedforward neural networks (FNN)

Feedforward neural networks process data in one direction, from the input node to the output node. A feedforward network uses a feedback process to improve predictions over time.

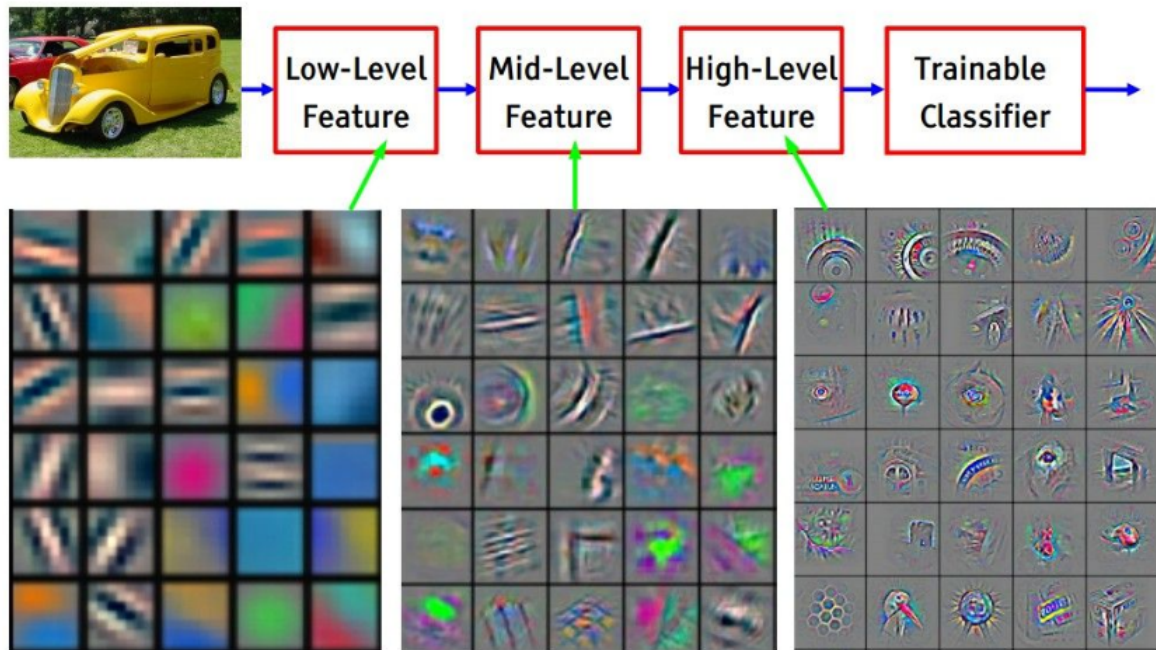


3.2 Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs) is a specialized type of deep learning architecture designed to excel at processing data with a grid-like structure, particularly images and videos. Unlike FNNs where each node in a layer connects to every node in the next, CNNs

introduce specific layers that exploit the spatial relationships within the data. It includes convolutional layers that apply filters to the input, pooling layers for down-sampling, and fully connected layers.

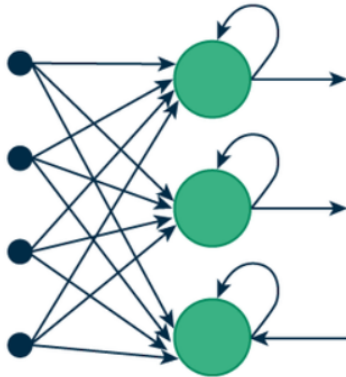
CNNs are used in Image recognition, Computer vision, Natural language processing,...



3.3 Recurrent Neural Network (RNN)

Recurrent Neural Networks (RNNs) are specifically designed to handle sequential data, where the order of information matters. Imagine an FNN trying to predict the next word in a sentence; it wouldn't consider the context of previous words. RNNs address this limitation by incorporating a memory of past information.

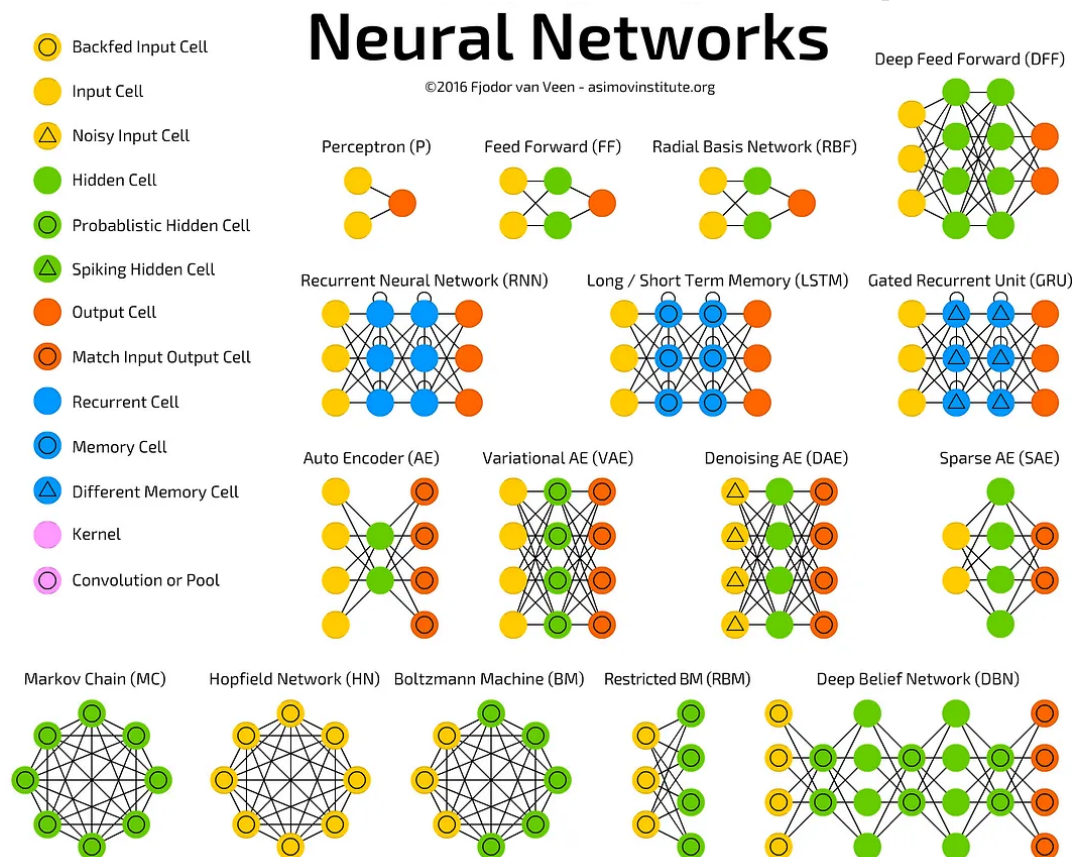
RNNs are used in Language processing, Time series forecasting, Music generation,...



3.4 Many more neural network models

Long Short-Term Memory (LSTM) Network, Gated Recurrent Unit (GRU) Network, Transformer Network,...

There can be many uses of neural network concept tailored to specific purposes. This makes Artificial Neural Network a cornerstone of modern AI development.



Chapter 4

Deep learning in neural networks

Deep learning in neural networks refers to the field of machine learning where neural networks are composed of multiple layers, allowing them to learn intricate representations of data. This capability enables the processing and understanding of complex tasks such as image recognition, natural language processing, autonomous driving, and more.

Deep neural networks (DNNs) consist of multiple layers of neurons, hence the term "deep". These layers can include hundreds or even thousands of neurons, enabling the model to learn increasingly complex representations of data.

Deep learning models have the ability to automatically learn high-level features from raw data. This eliminates the need for manual feature extraction and selection, as required in traditional machine learning methods.

Deep learning has contributed to the development of powerful neural network architectures such as Convolutional Neural Networks (CNNs) for image processing and recognition, and Recurrent Neural Networks (RNNs) for sequence data processing.

Despite its successes, deep learning faces challenges such as understanding the learning process, ensuring the transparency of decisions made by models, and mitigating overfitting in complex models.

Example for deep learning:

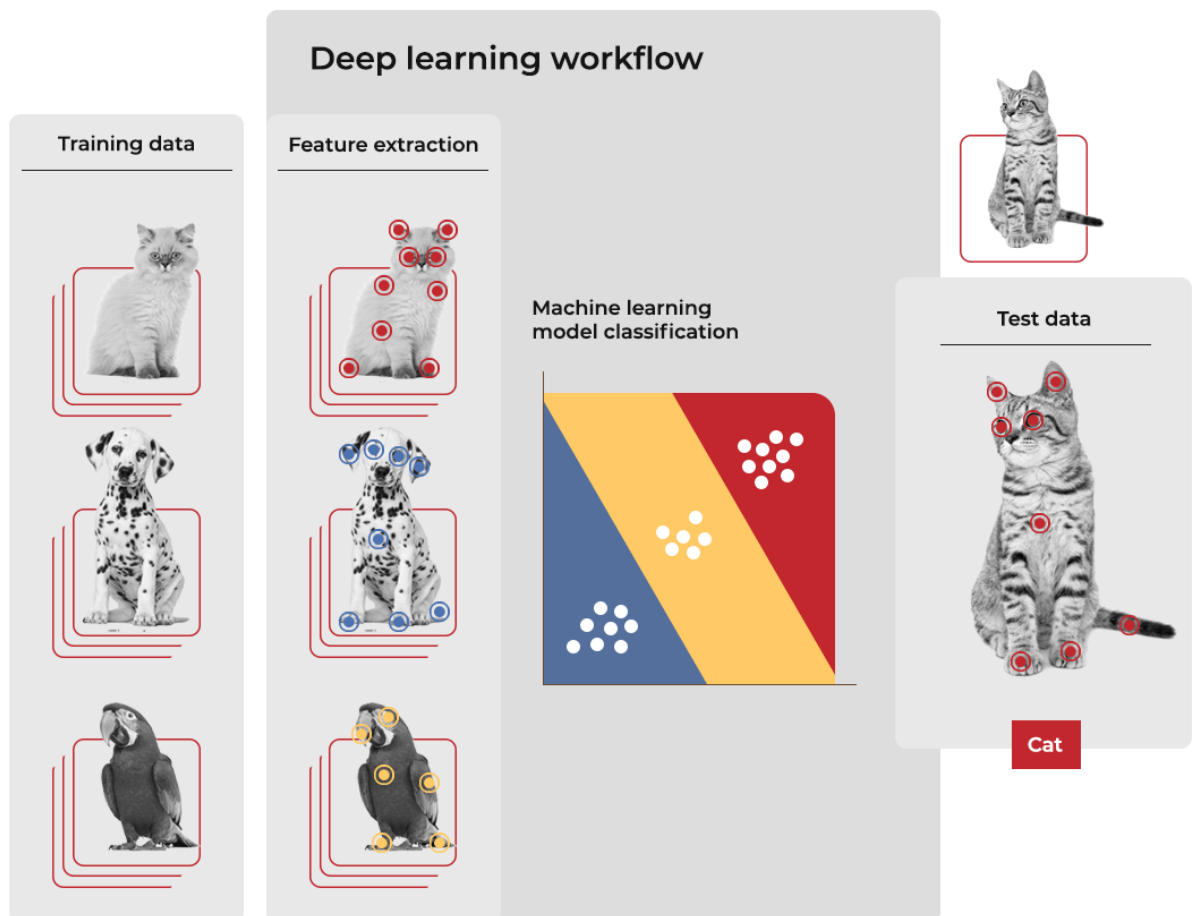
If you were training a machine learning software to identify an image of a pet correctly, you would need to take these steps:

- Find and label thousands of pet images, like cats, dogs, horses, hamsters, parrots, and so on, manually.
- Tell the machine learning software what features to look for so it can identify the image using elimination. For instance, it might count the number of legs, then check

for eye shape, ear shape, tail, fur, and so on.

- Manually assess and change the labeled datasets to improve the software's accuracy. For example, if your training set has too many pictures of black cats, the software will correctly identify a black cat but not a white one.

In deep learning, however, the neural networks would process all the images and automatically determine that they need to analyze the number of legs and the face shape first, then look at the tails last to correctly identify the animal in the image.



Chapter 5

Summary

Benefits:

- Can be applied to a wide range of tasks.
- Capable of achieving high levels of accuracy, especially with large datasets.
- Can learn and improve over time as they are exposed to more data.

Challenges:

- Designing and training neural networks can be complex and computationally intensive.
- Require large amounts of labeled data for training.
- Often considered "black boxes," making it difficult to understand how they make decisions.

Neural networks are powerful tools in AI, capable of learning from data to perform complex tasks. Their layered structure, ability to learn and adapt, and wide range of applications make them integral to many modern technologies.

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